

## Mechanics WS 2

- The velocity  $v$  of a particle moving along the  $x$ -axis is given by  $v^2 = 12 + 8x - 4x^2$ , where  $x$  is the distance from the origin  $O$ . Prove that the motion is SHM and find
  - the amplitude
  - the centre of oscillation
  - the period.
- A stone is projected with an initial velocity  $v_1$  in a vertical plane at an angle  $\theta$  to the horizontal and hits the ground at a point  $A$ . Another stone is projected with an initial velocity  $v_2$  at the same angle of projection, but  $v_2 > v_1$ . Let  $B(x_2, y_2)$  and  $C(x_1, y_1)$  be two points on the respective paths of flights at the same time  $t$ . Show that the gradient of segment  $BC$  is independent of time  $t$ . When the second stone just clears a wall of height  $h$ , the first stone hits the ground at  $A$ . If the wall stands at point  $D$  on the level ground, prove that  $AD = h \cot \theta$ . Further show that  $\tan(-\varphi) = \tan \theta - \frac{gT}{v_2 \cos \theta}$ , where  $\varphi$  is the angle made by the downward flight of the faster stone with the horizontal, and  $T$  is the time of flight for the slower stone. Hence show that  $v_2(\tan \theta + \tan \varphi) = 2v_1 \tan \theta$ .
- The nozzle of a water hose is at a point  $O$  on the horizontal ground. The water comes out of the nozzle with speed  $U$  m/s. Neglecting the air-resistance, prove that the water can reach the wall at a distance  $d$  from  $O$ , if  $U^2 > gd$ , where  $g$  is the acceleration due to gravity. If  $U^2 = 4gd$ , also prove that the maximum height that can be reached by the jet on this wall is given by  $\frac{15d}{8}$ .
- A body is projected with speed  $U$  from a height  $h$ , above a horizontal plane, at an angle  $\theta$  to the horizontal. Show that the range  $R$  on the horizontal is given by  $gR^2 \sec^2 \theta - 2U^2 R \tan \theta - 2hU^2 = 0$ . Further show that the maximum range  $R_1$  is given by  $R_1 = \frac{U}{g} \sqrt{U^2 + 2hg}$  and the corresponding angle of projection is given by  $\tan \theta = \frac{U^2}{gR_1}$ . Hence prove that  $h = R_1 \cot 2\theta$ .
- A particle is projected from a point  $O$  with velocity  $v$  at an angle  $\theta$  to the horizontal. It passes through the point  $P(x, y)$  in the vertical plane through  $O$  where  $(x, y)$  are the co-ordinates of  $P$  with respect to the rectangular axes at  $O$ . Prove that  $y = x \tan \theta - \frac{gx^2 \sec^2 \theta}{2v^2}$ . If  $x = 20$  m,  $y = 10$  m,  $g = 10$  m.s<sup>-2</sup>,  $v = 20$  m/s, find the two values of  $\tan \theta$  and using,  $t = \tan \theta$  and  $\sin 2\theta = \frac{2t}{1+t^2}$  prove that the ratio of two ranges is  $\frac{5}{3}$ .
- A particle is projected with velocity  $v$  at an angle  $\theta$  to the horizontal. Show that by suitable choice of axes, the equation of the path of the projectile is  $y = x \tan \theta - \frac{gx^2 \sec^2 \theta}{2v^2}$ . Prove the following:
  - There are two possible directions of projection given by  $\tan \theta_1$  and  $\tan \theta_2$  for a given range  $R$ .
  - $\tan \theta_1 + \tan \theta_2 = \frac{2v^2}{gR}$  and  $\tan \theta_1 \tan \theta_2 = 1$
  - Let  $T_1$  and  $T_2$  be the times of flights corresponding to angles of projections  $\theta_1$  and  $\theta_2$ . Prove that  $\frac{T_1}{T_2} = \frac{\sin \theta_1}{\sin \theta_2}$ .
  - From (b) prove that  $\theta_1 + \theta_2 = 90^\circ$
  - $R = R_1 \sin 2\theta$ , where  $R_1$  is the maximum range.
  - If  $\frac{T_1}{T_2} = \frac{2}{1}$ , then  $\frac{R'}{R_1} = \frac{4}{5}$
- A parachutist of mass  $m$  falls to ground from a plane. Given that air resistance is proportional to the square of his speed  $v$ :
  - Draw a diagram showing clearly the forces acting on the parachutist during his free fall
  - Deduce that  $\frac{d}{dx}(v^2) = 2g - 2kv^2$

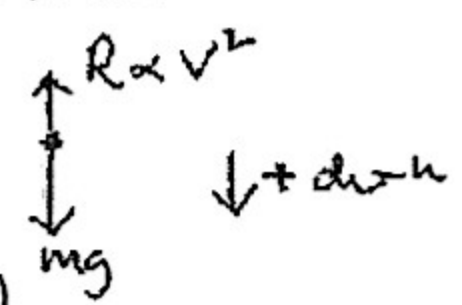


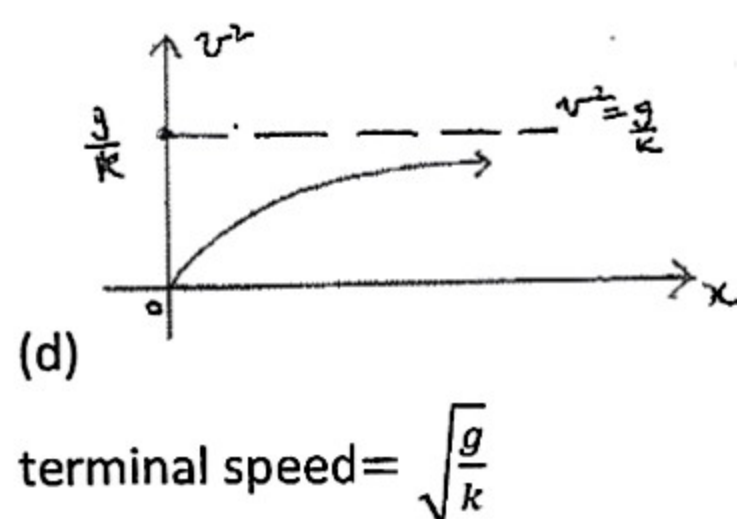
- c. Show that  $v^2 = \frac{g}{k} - Ae^{-2kx}$  satisfies the differential equation in part ii. and show that  $A = \frac{g}{k}$
- d. Sketch the graph of  $v^2$  against  $x$  and find an expression for the terminal speed of the parachutist during his free-fall.
8.  $P, Q$  and  $R$  are three points on the  $x$  - axis such that  $PQ = QR = 2$  m. A particle performs a SHM along the  $x$  - axis and is observed to have the velocities of 12, 10 and 6 m/s at  $P, Q$  and  $R$  respectively. Taking the origin at the centre of oscillation, find
- the constant  $n$  in the equation  $\ddot{x} = -n^2x$
  - the distance of  $P$  from the centre
  - the amplitude
9. A particle moves in a straight line and its position in metres at any time  $t$  seconds is given by  $x = 3 \cos 2t - 4 \sin 2t$
- By expressing the motion in terms of  $A \cos(nt + \alpha)$ . Show that the motion is simple harmonic.
  - Find the particle's greatest speed.
10. The speed  $V$  (m/s) of a point moving along the  $x$  axis is given by  $v^2 = 144x - 36x^2 + 180$
- Prove that the motion is SHM.
  - What is the maximum speed of the motion?
11. A particle is moving along the  $x$  -axis so that its acceleration after  $t$  seconds is given by  $\ddot{x} = -e^{-\frac{x}{2}}$

The particle starts at the origin with an initial velocity of 2 cm/sec

- If  $v$  is the velocity of the particle, find  $v^2$  as a function of  $x$
  - Show that the displacement  $x$  as a function of time  $t$  is given by  $x = 4 \log_e \left( \frac{t+2}{2} \right)$
12. A particle is moving in a straight line. At time  $t$  seconds it has displacement  $x$  metres from a fixed point  $O$  on the line, velocity  $v \text{ ms}^{-1}$  given by  $v = -\frac{1}{8}x^3$ , and acceleration  $a \text{ ms}^{-1}$ . The particle is initially 2 metres to the right of  $O$ .
- Show that  $a = \frac{3}{64}x^5$
  - Find an expression for  $x$  in terms of  $t$ .
  - Describe the limiting motion of the particle.
13. A particle moving on a straight line has acceleration  $a = -\frac{4}{x^2}$ . If  $v = 3$  when  $x = 1$ , then:
- Find  $v^2$  as a function of  $x$ .
  - Describe the motion.
14. A particle is moving on a straight line. At time  $t$  second it has displacement  $x$  metres from a fixed point  $O$  on the line, velocity  $v \text{ ms}^{-1}$  and acceleration  $a \text{ ms}^{-2}$ . The particle starts from  $O$  and at time  $t$  seconds  $v = (1 - x)^2$ .
- Find an expression for  $a$  in terms of  $x$ .
  - Find an expression for  $x$  in terms of  $t$ .

#### Answers:

- (a) 2 (b) 1 (c)  $\pi$
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- $v^2 = u^2 - 2ug \sin \theta + g^2 t^2$
- 
- $45^\circ, 71.57^\circ$
- (a) 



- (a)  $\sqrt{\frac{5}{2}}$  (b) 3.4 (c) 8.32
- (b)  $|v|_{\max} = 10 \text{ m/s}$
- (b) 18 m/s

- a.  $v^2 = 4e^{-\frac{x}{2}}$
- b.  $x = \frac{2}{\sqrt{1+t}}$ ; c. As  $t \rightarrow \infty$ , the particle is moving left, approaching  $O$  and slowing down at an ever decreasing rate with speed approaching 0.
- a.  $v^2 = \frac{8}{x} + 1$ ; b. particle moves to right with decreasing speed
- a.  $a = -2(1 - x)$ ; b.  $x = 1 - \frac{1}{t+1}$ ; c.  $t = 9 \text{ s}$